

Efficacy of Qingfei Paidu Granules combined with non-drug traditional Chinese medicine therapy in the treatment of patients with asymptomatic coronavirus disease

A retrospective study

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Abstract

The aim of this study was to retrospectively analyze the clinical efficacy and safety of Qingfei Paidu granules (QFPG) combined with non-drug therapy of traditional Chinese medicine (TCM) in treating patients with asymptomatic coronavirus disease 2019 (COVID-2019). A total of 450 patients admitted to the wards of Chongming Changxing Island isolation site from April 1 to 18, 2022 were retrospectively analyzed and divided into 3 groups according to the treatment that they received. W1 was not treated (control group), and W2 was given QFPG (TCM treatment), W3 (TCM combined with TCM non-drug therapy) was treated with QFPG, five-element music therapy and Gong. We retrospectively analyzed the medical history data, including general information, time of first turning negative, length of hospital stays, number of cases classified as mild or ordinary, number of cases with adverse events/adverse reactions. The conversion to negative time of the 3 groups was 6.50, 4.40, and 3.81 days, respectively, and there were significant differences among the 3 groups ($P < .001$). The hospital duration in the 3 groups was 8.45, 5.72, and 5.53 days, respectively, and there were significant differences among the 3 groups ($P < .001$). The number of adverse events W1 (5/150, 3.3%), W2 (8/150, 5.3%), and W3 (4/150, 2.7%) was consistent among the 3 groups ($P > .05$). QFPG with TCM non-drug therapy is effective in patients with asymptomatic COVID-2019, which can effectively shorten the time to double negative and reduce the proportion of patients with mild disease.

Abbreviations: TCM = traditional Chinese medicine, QFPG = Qingfei Paidu granules.

Keywords: COVID-19, five-element music therapy, Gong, Qingfei Paidu granules (QFPG), traditional Chinese medicine

1. Introduction

Novel coronavirus pneumonia, referred to as “COVID-19,” is an acute respiratory infectious disease caused by a newly emerged coronavirus type 2 (SARS-COV-2). The disease is characterized by fever, fatigue, and a dry cough. In severe cases, it can rapidly progress to acute respiratory distress syndrome, septic shock, and difficult-to-correct metabolic acidosis and coagulation disorders.^[1] Since the outbreak of COVID-19 in December 2019, in China it has caused a pandemic that has lasted for more than 2 years, during which the virus has continuously mutated, forming numerous strains including Alpha, Beta, Gamma, Delta, and Omicron.^[2,3] Asymptomatic COVID-19-infected individuals (hence

referred to as asymptomatic individuals) exhibit no significant clinical symptoms such as fever, cough, sore throat, or other self-perceived or clinically recognizable symptoms and signs, but positive in the novel coronavirus test of respiratory tract specimens. Asymptomatic individuals can be divided into 2 situations: one is an infected person who has a positive nucleic acid test and has no self-perceived or infected person is positive, and there is no self-perceived or clinically identifiable symptoms and signs at the time of sampling, but then some clinical manifestation occurs, that is, asymptomatic infection in the incubation period.^[4]

Omicron is characterized with high infectivity and resulted in frequent outbreaks throughout China, including the 2022 outbreak in Shanghai. An Omicron infection is characterized by high

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The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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occult symptoms, increased immune escape ability, and asymptomatic and mildly symptomatic infected patients, creating challenges to epidemic prevention and control. Traditional Chinese Medicine (TCM) has been helpful in the prevention and treatment of COVID-19. In the *TCM Diagnosis and Treatment Protocol for COVID-19 (Trial Edition 9)*, Qingfei Paidu granules (QFPG) are used to treat mild, typical, and severe confirmed COVID-19 cases.^[5] QFPG has also been used to treat asymptomatic infected individuals and mild cases in the *Implementation Plan for the Application of Traditional Chinese Medicine in Shanghai Novel Coronavirus Square Cabin*. Non-drug treatment of TCM has been added to the *Consensus of Experts on the Diagnosis and Treatment of Novel Coronavirus Infection in Shanghai (Spring 2022 Edition)*. This TCM non-drug treatment includes acupuncture, TCM emotional therapy, TCM guidance and other techniques, five-element music therapy, acupoint massage, ear sticking and other treatments.

According to the characteristics of the epidemic, this study used TCM and TCM non-drug therapy to intervene in infected patients to avoid the transformation of asymptomatic infected patients and exacerbation of mild patients at the onset of COVID-19 infection, to provide basis and reference for the clinical treatment of asymptomatic novel coronavirus infected patients.

2. Methods

2.1. Participants

The data of 450 asymptomatic COVID-19-infected persons admitted to 3 wards of Chongming Changxing Island isolation and treatment shelter site, from April 1, 2022, to April 18, 2022, were collected. Participants included patients diagnosed with asymptomatic COVID-19 infection and referred to Chongming Changxing Island hospital within 48 hours of the first novel positive Coronavirus nucleic acid test and were at least 18 years of age. There was no gender limitation, and all participants conformed to TCM “heat syndrome.” Patients were excluded from the study if they presented with other infections, tumors, autoimmune diseases, and other serious diseases. Pregnant and lactating women were also excluded as well as cases where complete medical information was not available. This study was approved by the Ethics Committee of Shanghai Municipal Hospital of Traditional Chinese Medicine.

Participants were divided into 3 groups according to different treatment methods: control group (W1), QFPG treatment group (W2), and QFPG combined with TCM non-drug therapy group (W3).

2.2. Treatment protocols

The control group (W1) were given no medical treatment. The TCM treatment group (W2) were treated with QFPG, made-up of Ephedra, prepared Glycyrrhiza, almond, raw gesso, cassia twig, Alisma rhizoma, tractylodes atractylodes, Poria cochue, Bupleurum, Scutellaria baicalensis, Pinellia ginger, ginger, Aster, winter flower, Belamcanda chinensis, Asarum, Chinese yam, Immaturus aurantii, tangerine peel, and Pogostemon cablin. QFPG was provided by Shanghai Hospital of TCM in accordance with the Medication plan for asymptomatic infected persons, described in Implementation plan for Application of traditional Chinese medicine in coronavirus capsules in Shanghai. The QFPG was taken with boiled water 40 minutes after a meal, 200 mL at a time, twice a day. The TCM plus TCM non-drug therapy treatment group, W3, were treated with QFPG as for W2 in addition to TCM non-drug therapy. W3, TCM with TCM non-drug therapy included QFPG as described with five-element music therapy and gong therapy. The protocol included “Three Stanzas of Plum Blossoms” played at 7:00, “The Spring Snow” and “Yangguan Triassic” played at 15:00, “Frontier Song” played at 19:00, and “Daughter’s Love” is played at 21:00, each

day. Traditional Gong therapy happened at 16:00 every day, with an 8-section brocade and Shi’s twelve-character health preserving exercises, led by an exercise teacher.

2.3. Observation indicators

To compare the efficacy of the treatment groups, the time taken to reach a negative COVID-19 nucleic acid test was recorded for each participant. Referred to as “conversion time,” this was measured as 2 consecutive negative COVID-19 tests taken at least 24 hours apart. The length of hospital stays, from admission to discharge, was also recorded and the conversion rate of asymptomatic COVID-19 infection to mild or common symptoms. Physiological measurements were also recorded daily for each participant, including body temperature, heart rate, respiration, blood pressure, blood oxygen saturation and whether there are adverse events or reactions during treatment.

2.4. Discharge criteria

Participants with two consecutive coronavirus nucleic acid assays (fluorescent quantitative PCR method) taken at least 24 hours apart with a CT value < 35 or ≥ 35 for both the N and ORF and a cutoff value of 40.

2.5. Statistical analysis

SPSS 26.0 (IBM Corp., Armonk, NY) software was adopted for statistical analysis. Viral double time of negative transformation was analyzed by Kaplan–Meier method listing the quartiles of the transformation time and 95% confidence interval (CI) (Brookmeyer–Crowley method) and comparison between groups was made using a log rank test. Multiple comparisons were performed by one-way analysis of variance (ANOVA) or a Kruskal–Wallis test, with a Tukey–Kramer multiple comparisons post-test as a appropriate. The rate of the first conversion time of the virus during 7 days in each group and the 95% CI (with log-log transformation) were calculated. Kaplan–Meier curves were plotted for the rate of the first conversion time of the virus in each group. Length of hospital stay was analyzed using the Kaplan–Meier method, with quartiles and 95% CI, censoring rates, and comparisons between groups performed using the log rank test. Light or ordinary type case conversion rates among groups were explored using the chi square or Fisher’s exact probability method, the 95% CI calculated.

3. Result

The mean ages of patients in the W1 group, W2 group, and W3 group were 41.26 ± 13.52 , 41.18 ± 13.72 , and 41.87 ± 12.36 years, respectively ($P = .287$). There were 92 males and 58 females in the W1 group, 74 males and 76 females in the W2 group, and 80 males and 70 females in the W3 group ($P = .104$). There were no statistically significant differences among these 3 groups in terms of smoking, drinking, body mass index, past medical history, COVID-19 vaccine, and medication used (all $P > .05$) (Table 1).

3.1. Conversion time to double negative COVID-19 test

The conversion time to double negative COVID-19 test in W1, W2, and W3 groups was 2.81 days, 1.92 days, and 2.29 days respectively. The comprehensive comparison among multiple groups showed that the difference in the double negative time was statistically significant ($P < .001$) (Table 2). Compared with W1, W2, and W3 could shorten the time of turning to double negative ($P < .001$), and W2 and W3 also shortened the time of turning to double negative ($P < .001$) (Table 3). Figure 1 for the conversion rate from 0 to 7 days in each group.

Table 1**Comparison of general information.**

Variables	W1	W2	W3	χ^2/F test	P
N	150	150	150		
Age (yr)	41.26 ± 13.52	41.18 ± 13.72	41.87 ± 12.36	1.32	.287
Male (%)	92 (61.33)	74 (49.33)	80 (53.33)	4.519	.104
BMI (kg/m ²)	25.27 ± 3.26	25.39 ± 3.18	26.11 ± 3.36	1.582	.276
Smoking (%)	36 (24.0)	35 (23.3)	29 (19.33)	1.106	.575
Drinking (%)	62 (41.33)	59 (39.33)	64 (42.67)	0.349	.839
Past medical history (%)					
Bronchial asthma	2 (1.33)	3 (2.00)	1 (0.67)	1.014	.602
COPD	2 (1.33)	3 (2.00)	2 (1.33)	0.29	.865
Coronary artery disease	6 (4.00)	3 (2.00)	5 (3.33)	1.032	.597
Diabetes	8 (5.33)	14 (9.33)	11 (7.33)	1.766	.413
Hypertension	33 (22.00)	27 (18.00)	25 (16.67)	1.508	.47
Hyperlipidemia	10 (6.67)	11 (7.33)	8 (5.33)	0.516	.773
Others	21 (14.00)	22 (14.67)	22 (14.67)	0.036	.982
COVID-19 vaccine (times)	2.11 ± 0.95	2.13 ± 1.23	2.14 ± 1.18	0.594	.731
Medication used					
Oseltamivir	127 (84.67)	130 (86.67)	123 (82.00)	1.252	.535
Arbidol	49 (32.67)	47 (31.33)	53 (35.33)	0.562	.755
Macrolide antibiotics	36 (24.00)	37 (24.67)	44 (29.33)	1.317	.518
Ofloxacin	29 (19.33)	22 (14.67)	31 (20.67)	1.998	.368
Others	15 (10.00)	7 (4.67)	12 (8.00)	3.118	.21

Table 2**Conversion time to double negative COVID-19 test.**

	W1	W2	W3	Statistic	P value
N	150	150	150	218.619	<.001
Censored (%)	46 (30.7)	0 (0.0)	0 (0.0)		
Events (%)	104 (69.3)	150 (100.0)	150 (100.0)		
25% (95% CI)	5 (4, 6)	4 (3, 4)	3 (NA, NA)		
50% (95% CI)	7 (NA, NA)	5 (NA, NA)	4 (3, 4)		
75% (95% CI)	NA (NA, 7)	5 (NA, NA)	4 (4, 5)		

Table 3**Comparison of conversion time to double negative COVID-19 test.**

Group	Statistic	P value*
W1:W2	110.200	<.001
W1:W3	213.800	<.001
W2:W3	9.889	.005

*Tukey-Kramer was used to correct multiplicity comparisons.

3.2. Length of hospital stay and outcome

The Length of hospital stay of W1, W2, and W3 groups was 6.63 days, 5.26 days, and 5.29 days respectively. The difference of length of hospital stay was statistically significant ($P < .001$) (Table 4). Compared with W1, W2 and W3 could shorten the length of hospital stay ($P < .001$), but there was no significant difference between W2 and W3 ($P > .05$) (Table 5). The 0- to 7-day hospitalization rate of each group is shown in Figure 2.

The conversion rates of light/ordinary cases in W1, W2, and W3 groups were 15.3%, 6.7%, and 4.0%, respectively. There was no significant difference between the two groups ($P > .05$) (Tables 6 and 7).

3.3. Occurrence of adverse events

During the clinical study, the 3 groups showed no abnormalities in vital signs, including body temperature, heart rate,

respiration, blood pressure, oxygen saturation and other physiological indicators. Adverse events included nausea, diarrhea, and constipation, which after symptoms were treated, with no statistical significance ($P > .05$) (Table 8).

4. Discussion

In the TCM section of novel Coronavirus Diagnosis and Treatment Protocol for Pneumonia (Trial Edition 9), COVID-19 is defined as “epidemic disease” according to its clinical manifestations and pathogenic characteristics. “Epidemic disease” was first mentioned in *Huangdi neijing*, and *Su Wen Discussion on Acupuncture method* said, “The 5 epidemics are all easily infected, with no size and similar symptoms.”^[6] Zhongjing Zhang also recorded in Han Dynasty Treatise on Febrile Diseases: “Since Jianan’s era, less than ten years have passed, two-thirds of whom died and 10 lived with typhoid fever in their seventh,” and created a classic term for treating typhoid fever in a series, including apricot Shigan Decoction and so on.^[7] Jutong Wu of the Qing Dynasty said in *Wenbing Tiaobian*: “Warm epidemic, severe gas epidemic, and more dirty, so every family”, this shows the infectious character of plague.

According to TCM clinical observation from various regions in China, COVID-19 is mainly characterized by “dampness, heat, toxicity and deficiency.”^[8,9] The pathogenesis is “warm heat turbid poison,” “cold dampness” and other pathogenic factors invade the human body, and the disease is in the lung.^[10] The core pathogenesis of COVID-19 is therefore dampness, heat, toxin, blood stasis and deficiency, and dampness toxin impingement in the lung, obstructing lung and occluding lung

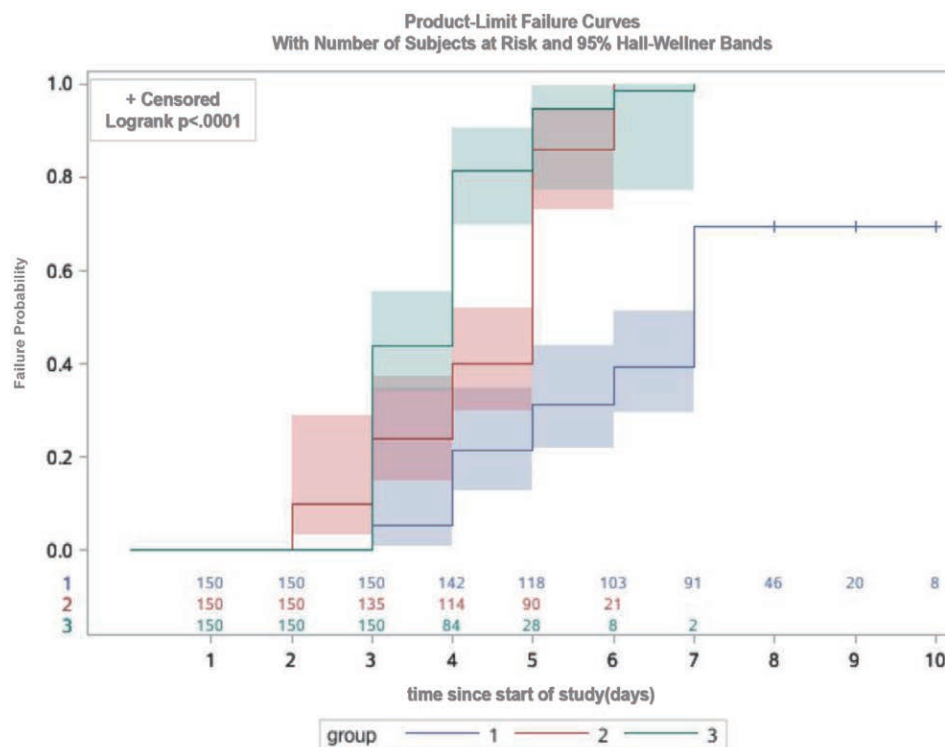


Figure 1. K-M plot of the conversion time to COVID-19 negative for each treatment group.

Table 4

Length of hospital stay.

	W 1	W 2	W 3	Statistic	P value
N	150	150	150	219.422	<.001
Censored (%)	100 (66.7)	0 (0.0)	7 (4.7)		
Events (%)	50	150	143		
25% (95% CI)	7 (6,7)	6 (4,6)	5 (NA, NA)		
50% (95% CI)	NA (NA, NA)	6 (NA, NA)	5 (5,6)		
75% (95% CI)	NA (NA, NA)	6 (6,7)	6 (NA, NA)		

Table 5

Post hoc analysis of the variation in length of hospital stays among treatment groups.

Group	Statistic	P value*
W1:W2	140.8	<.001
W1:W3	200.2	<.001
W2:W3	2.3707	.272

*Tukey–Kramer correction for multiple comparisons was applied.

run through the whole disease process.^[11] As such, TCM treatment emphasizes light to heavy stage dialectical treatment. Spleen and stomach are the hinge of pathogenesis evolution of wet toxin syndrome.

From the perspective of 6 channels transmission, Chen Rui et al^[12] believed that the evil of dampness toxin trapped the spleen and closed the lung at the initial stage, then entered the body to transform heat into Yang Ming, and formed the syndrome of Yang Ming Fu-fu, dampness toxin and stasis heat closing inside, and heat deep convulsion deep. Most of the patients are the current phase of damp-heat diseases, and the severe symptoms of internal closure and external detachment

are the changes of the disease. Sichao Zhang et al^[13] analyzed from the perspective of the blood transmission of Weiqi, that the wet toxicity was relatively severe at the beginning of the disease, and most of the disease was associated with Weiqi, then developed into qi syndrome. Wet poisons quickly dry, can be introduced into the camp blood, serious pulmonary source to eliminate the crisis.

The *National Program* recommends that QFPD can be applied to all types of patients in the clinical treatment period. The formula has been optimized and combined from several classical formulae for external febrile diseases caused by cold pathogens in *Treatise on Febrile and Miscellaneous Diseases*.^[14] It can be regarded as a combination of Ephedra decoction and Wuling Powder, which not only treats cold closure, but also helps relieve urine and dispel dampness. Since lung closure is more serious than an asthma cough, it can also be combined with Shigan Ephedra decoction and Orange Trifoliate Ginger Decoction. Xiaochaihu decoction is designed for shaoyang disease, with half surface and half li and tongli sanjiao, which not only prevents evil from getting into the body, but also regulates liver and stomach, and protects digestive function.^[15] In this compound, many herbs go to the spleen, stomach, heart and kidney channels at the same time, and play a protective role in each viscera.^[16]

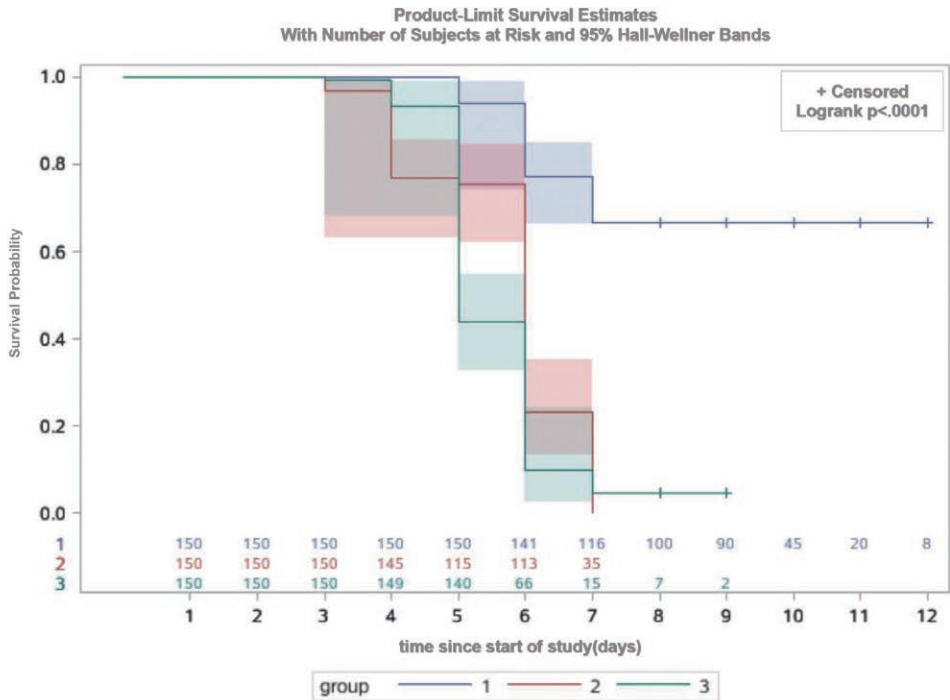


Figure 2. K-M plot of length of hospital stay.

Table 6

Mild or common forms of case conversion.

	W1	W2	W3
Conversion to less severe disease (%)	23 (15.3)	10 (6.7)	6 (4.0)

Table 7

Between group comparisons of case conversion rates for mild or common forms.

Group	Statistic	P value	Method
W1:W2	0.424	.515	Chi square
W1:W3	2.753	.097	Chi square
W2:W3	1.056	.304	Chi square

Table 8

Occurrence of adverse events.

Group	N	Nausea	Diarrhea	Astriaction
W1	150	0	1	4
W2	150	3	3	2
W3	150	2	1	1

Ba Duan Jin is a treatment method with national characteristics in the field of traditional rehabilitation medicine in China. Its health preservation concept is in the same line with the basic theory of TCM. It is a kind of exercise mode that takes the exercise of essence, qi and spirit as the core and uses different skills and movements to adjust the functions of zang-fu organs based on syndrome differentiation. Its movement is simple, easy to learn, is low intensity aerobic exercise, in the prevention and treatment of respiratory system diseases and immune diseases has a certain benign effect, is one of the important methods of

TCM “cure of disease.” Before the imbalance of zang-fu organs, the “cure of disease” of TCM consciously regulates the functions of all organs in time, so as to restore the balance state, so that qi and blood can run smoothly, and block the environmental basis for the generation of disease from the source, so as to prevent the infection and spread of disease. Therefore, through the practice of Ba Duan Jin, prevention before disease can strengthen the body’s constitution, enhance the ability to resist disease evil, so that there is no chance for disease evil to enter, and can achieve the role of “prevention before disease” in the prevention and treatment of COVID-19. The 5-element music therapy is based on the theory that the 5 tones enter the 5 zang organs and the 5 elements produce restraint. Through the music style and rhythm, it plays a resonance effect in the human body and promotes the coordination of qi, blood, body fluid and zang-fu organs functions through conduction, so as to finally achieve the purpose of calming the mind. Among the music selected in this study, the tune music can enter the heart meridian and small intestine meridian, further promote the patient’s qi unblock and regulate the circulatory system through soothing and soft melody. “Shi’s twelve word life cultivation skill” is a guidance method created by Professor Qi Shi, master of TCM and the fourth generation successor of Shanghai Shi’s injury department, based on decades of clinical experience and scientific research experience, and inheriting the martial arts essence of Ziping Wang, master of martial arts injury department. It has the effect of health preservation and health care. This method follows the meaning of guiding qi to make harmony and pulling body to make softness, combining movement, breathing and mind, aiming to achieve the functions of qi and blood circulation, relaxing tendons and activating collages and regulating body and mind through reasonable limb joint activities and breathing movement and emotional conditioning. This method includes 12 joint movements including “washing face,” “combing hair,” “rubbing ear,” “rubbing neck,” “loosening neck,” “pressing waist,” “turning waist,” “kneading knee,” “squatting hip,” “rubbing sanjiao,” “exhuming old and absorbing new,” and “regulating limbs.” Through the whole-body movement, it can improve the regulating ability of the nervous system of the body, adjust humoral factors and metabolism. It can improve the cardiopulmonary

function, enhance the peristalsis and digestive capacity of the gastrointestinal system, effectively relieve pressure, remove anxiety, depression, and other adverse emotions, so as to promote the adjustment of physical and mental health.

Our results showed that the time of double negative conversion in W3 group was significantly lower than that in the other two groups, and the length of hospital stay was significantly lower than that in W1 group, slightly lower than that in W2 group, and the number of cases of mild conversion rate was also lower than that in the other two groups. Many cases of adverse events occurred 3 groups there was no statistical significance prompt TCM QFPD combined with non-drug therapy has a curative effect on asymptomatic COVID-19 infection. The TCM combination treatment effectively shortened recovery time, and reduce the proportion of patients with mild outcomes, for will be coronavirus has certain guiding significance to the treatment of asymptomatic infections. Due to the small number of cases at present, there may be differences between the two groups and the statistical significance was not achieved. However, the research results show that the combined treatment group of TCM plus TCM non-drug therapy has a positive trend towards curing COVID-19.

Limitations should be addressed: this study is a retrospective study on the novel Coronavirus asymptomatic infected patients confirmed and discharged from the isolation treatment site of Chongming Changxing Island. At present, the case data are limited, and the sample size is small, so there may be some data deviation, and more clinical data and materials are needed to prove the exact clinical efficacy.

TCM QFPG joint non-drug therapy for treatment of asymptomatic COVID-19 infections demonstrate a clinical curative effect and treatment. This is indicated by the faster conversion time to a double negative COVID-19 test of patients in the group with both TCM approaches compared to the control and the TCM QFPG treatment. The duration of hospital stay for participants in the joint TCM group was also shorter and time to mild or ordinary outcome was also lower. The number of cases in the treatment of asymptomatic infection has important clinical value.

Author contributions

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References

- [1] Majumder JMT, Minko T. Recent developments on therapeutic and diagnostic approaches for COVID-19. *AAPS J.* 2021;23:14.
- [2] WHO. Tracking SARS-CoV-2 variants. EB/OL. 2021.
- [3] Chun Gao FF, Jingjing J, Xiaowen Y, et al. Recent research progress on novel coronavirus variant“ Omicron “. *J Hainan Med Univ.* 2022;28:481–5.
- [4] What are novel coronavirus asymptomatic infected persons. *Chin J Health Inf Manage.* 2020;17:162.
- [5] National Health Commission of the People's Republic of China. Available at: <http://pmo433a4d.pic15.websiteonline.cn/upload/u6yx.pdf> [access date October 13, 2022].
- [6] Jin GWJ. Preliminary understanding of COVID-19 in Traditional Chinese Medicine. *Tianjin Traditional Chinese Medicine.* 2020;37:247–50.
- [7] Ruibing Lv WWXL. Clinical observation of Lianhua Qingwen granule combined with conventional western medicine in the treatment of 63 suspected COVID-19 cases Chinese. *J Tradit Chin Med.* 2020;61:655–9.
- [8] Wenke Zheng JZ, Fengwen Y, Yuguang W, et al. A comprehensive analysis of the diagnosis and treatment of COVID-19 by traditional Chinese medicine. *Chin J Trad Chin Med.* 2020;61:277–80.
- [9] Ping Yue ST, Huan Y, Fang W, et al. Pathogenesis and formulation of traditional Chinese medicine (TCM) prevention and treatment for COVID-19. *Chin J Exp Formulae.* 2020;26:13–9.
- [10] Jing Chen WW, Chunyang S, Jianguo F. Thoughts on the prevention and treatment of COVID-19 with traditional Chinese Medicine. *Chinese Herbal Medicine.* 2020;51:1106–12.
- [11] Jia-ju Ma MCY-gW. Overview of TCM syndrome treatment of COVID-19. *Beijing Trad Chin Med.* 2020;39:95–101.
- [12] Rui Chen YL, Xunhua X, Qing M, et al. A preliminary study on the treatment of 52 cases of COVID-19 with traditional Chinese medicine and analysis of typical case records. *Chin J Trad Chin Med.* 2020;61:741–4.
- [13] Sichao Zhang Z-JY, Xing-cai Z, Yuxia M, et al. Differentiation and treatment of COVID-19 in Traditional Chinese Medicine Shandong. *J Tradit Chin Med.* 2020;39:315–9.
- [14] Research on screening effective prescriptions of traditional Chinese medicine has made phased progress – Clinical observation in pilot provinces shows that the total effective rate of Qingfei Jiedu Decoction can reach more than 90%. *Bulletin of Traditional Chinese Medicine.* 2020;19:9.
- [15] Boshou Xue KYYX. A theoretical analysis of the rapid and effective treatment of COVID-19 with Qingfei Dedu Decoction. *Chin J Trad Chin Med.* 2020;61:461–2.
- [16] Jing Zhao ST, Jian Y, Jianfeng L, et al. The mechanism of Qingfei Jiedu Decoction in the treatment of COVID-19. *Chin Herbal Med.* 2020;51:829–35.